

Bus Access in Montgomery County

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Bus access is important to everyone. For many people, a car is a financial burden; many families cannot afford to replace their car if something happens to it, and many people live paycheck to paycheck to pay for their cars. The cost of transportation exacerbates poverty and financial struggles, and it does not have to be this way. Investing in our public buses to increase access to people around the county lessens the burden of transportation for many people and provides an alternative transportation method for anyone who can't or doesn't want to drive, and anyone who finds themselves temporarily without a car. Traffic is also a major issue and it is well known that widening streets does not solve it; instead, going to the root of the issue and reducing the number of cars on the road can dramatically improve traffic (Anciaes et al., 2025), which effective bus service can easily achieve. Less cars on the road means less traffic, less emissions, and safer streets.

Montgomery County, Maryland, is served by two major bus agencies, Washington DC's WMATA Metrobus and MCDOT Ride On. In this project, I look at what bus access in Montgomery County looks like in data, examine relationships between measures of bus access and vehicle use, and look for areas in the county that my analysis suggests would benefit from improvement to existing service.

Data used

This project uses data from the U.S. Census Bureau, WMATA, MCDOT, the Montgomery County Planning Department, Open Data DC, and dataMontgomery. This includes TIGER/Line Shapefiles of 2025 census tracts, 2024 American Community Survey 5-year estimates for: Household Size by Vehicles Available, Number of Workers in Household by Vehicles Available, and Demographic and Housing Estimates. The data from Open Data DC is Metrobus stops and lines shapefiles, and the data from WMATA and MCDOT include General Transit Feed Specification (GTFS), from the WMATA API and the MCDOT website. The start date for the MCDOT Ride On GTFS was January 11, 2026, and the end date was May 2, 2026; for WMATA Metrobus, the start date was March 16, 2026, and the end date was June 13, 2026. Bus data also includes average daily ridership from February 2026. The WMATA Metrobus ridership was provided by the WMATA website and the Ride On ridership was provided by DeRon Jordan, Senior Planning Specialist at Ride On Transit Services. The data from the Montgomery County Planning Department is a property file geodatabase, and from dataMontgomery, a map of communities in Montgomery County.

Variables and terms

In this report and project, there are several terms I use to refer to specific concepts. When I refer to “homes” or “dwellings”, I am referring to every individual residential unit, which counts apartment units and single-family homes individually. “Walking distance” is defined in this project as between 0 and ½ miles. When I refer to “cities”, I am referring to the map of communities from dataMontgomery.

Data ingestion

Many variables were created in ArcGIS and used for analysis and visualization in R. The property file geodatabase, a dataset of all buildings in Montgomery County, was filtered to “residential buildings” by selecting the first 16 land use codes: 111, 116, 114, 115, 118, 119, 112, 113, 117, 140, 121, 122, 123, 124, 125, and 126. As a note, this includes religious quarters, boarding houses, membership lodgings, and nursing homes. The buildings were joined to census tracts to create sums of all the homes in those tracts in walking distance of a bus route or line. GTFS data was used in ArcGIS to plot bus routes and stops, which were also joined to census tracts for analysis.

Findings (numbers and statistics)

Most census tracts have about 5 bus routes that travel through them, and most buses travel through about 12 census tracts. The most used bus week-round in the county is M60, with, on weekdays, 181 trips and an average of 6,602 riders in February 2026. The most used Ride On bus week-round is route 55 with, on weekdays, 139 trips and an average of 5,783 riders. 51.01% of households in the county have more people than vehicles; 13.93% have less vehicles than workers; and 8.5% have 0 vehicles altogether. Included is a map of census tracts colored by proportion with less vehicles than workers (Figure 1).

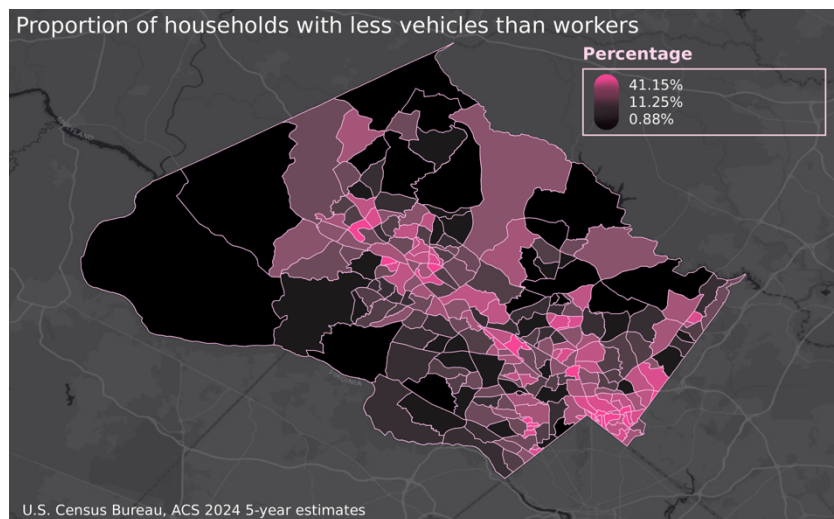
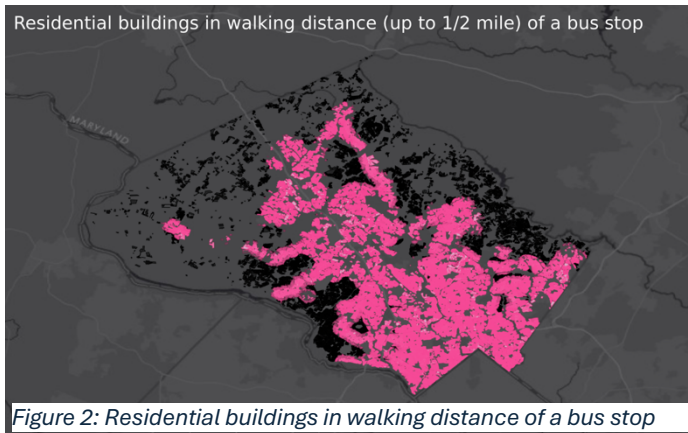


Figure 1: Proportion of households with less vehicles than workers, by tract

84.45% of homes in the county are within ¼ mile of a bus stop and 94.11% of homes are within ½ mile of a bus stop (Figure 2). 40.2% of the land in the county (in square miles) is within ½ mile of a bus stop, and only 25.98% within ¼ mile.



Despite the large proportion of homes in walking distance of a bus stop, only 13.93% of households have less vehicles than workers. Simply having a bus stop near their home doesn't mean that a person will use the bus; it isn't a good

measure of bus access or service by itself, and it does not imply that someone is less likely to use or own a vehicle.



Figure 3: Homes in walking distance of a bus stop, populous tracts

These statistics are illustrated by bar charts of the top 60 most populous census tracts, with the removal of one tract from Poolseville for outlying square mileage. The vast majority of homes in these tracts are in walking distance of a bus stop – most of them not further than ¼ mile (Figure 3). In most of these census tracts, the majority of the land is within walking distance of a bus stop as well (Figure 4). However, the overwhelming majority of households in these tracts have an



Figure 4: Land in walking distance of a bus stop, populous tracts

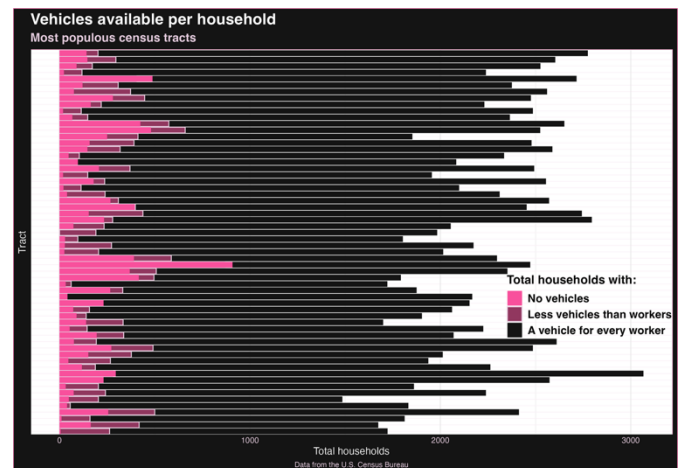


Figure 5: Vehicles available per household, most populous tracts

even number of people and vehicles, meaning that every single worker has their own vehicle (Figure 5).

There is considerable reduction in service from weekdays to weekends, which varies by census tract. Altogether, counting all bus trips for all routes across the county, there is a 33.56% reduction in trips from a weekday to Saturday, and a 41.58% reduction from a weekday to Sunday. However, there isn't a strong relationship between reduction of trips and proportion of households with less vehicles than workers.

Two variables that do have a relationship with proportion of households with less vehicles than workers are weekday trips per person (Figure 6) and bus routes per square mile (Figure 7). "Weekday trips per person" refers to the total number of trips on a weekday schedule divided by the population of a census tract. This essentially means the total number of buses driving through a census tract on a weekday, divided by population. When comparing this variable to proportion of households with less vehicles than workers, there is a correlation of 0.45 for census tracts, and 0.70 for census tracts grouped into cities. "Bus routes per square mile" refers to the total number of bus routes intersecting with a census tract, divided by population; this had a correlation of 0.54 with proportion of households with less vehicles when looking at census tracts, and a correlation of 0.80 when grouping to cities.

As a note, the colors on the "bus routes per square mile" map (Figure 7) are misleading due to outlying census tracts centered around the Paul S. Sarbanes Transit Center in Silver Spring.

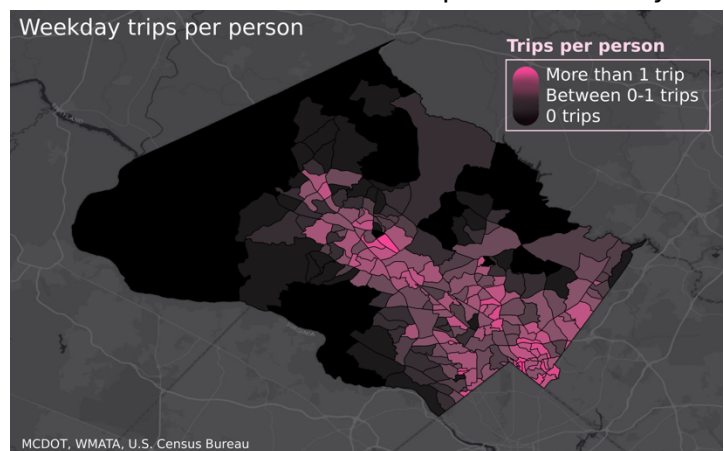


Figure 6: Weekday trips per person

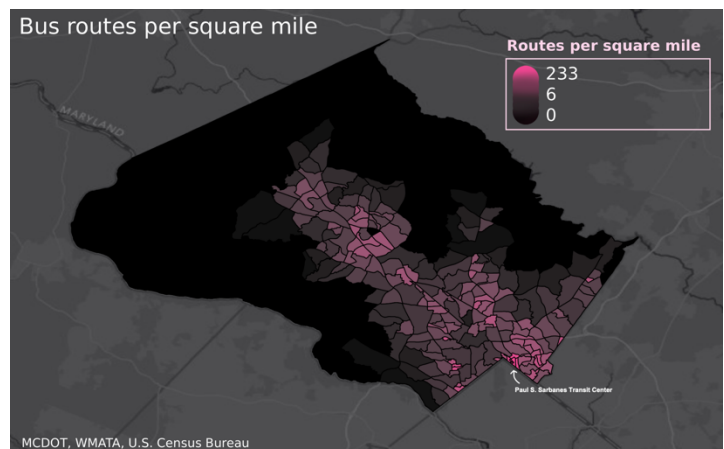


Figure 7: Bus routes per square mile

Statistical analysis

I use multiple linear regression to examine relationships between variables measuring bus access and the proportion of households with less vehicles than workers. I

found that there are many relationships with variables measuring bus access; some notable variables that have a positive relationship with the target are proportion of land ¼ mile from a bus stop, proportion of homes ¼ mile from a bus stop, bus routes per square mile and trips per person.

I performed regression with census tracts, and then with census tracts grouped into cities. The latter is not a perfect reflection of the cities themselves. It represents the data from different groups of census tracts combined, which reflect patterns in the general shape of cities/communities in the county.

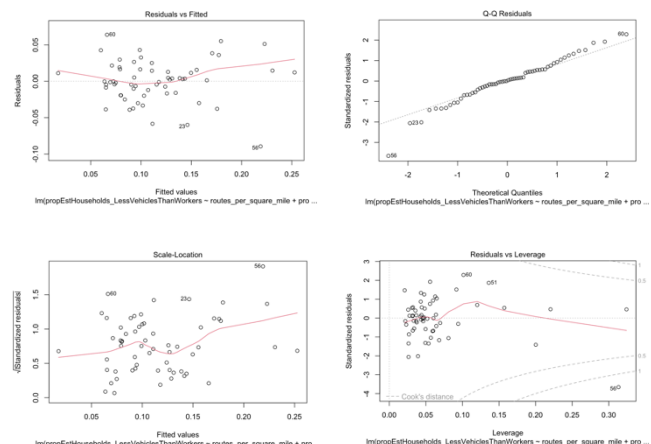
Regression with census tracts grouped into cities produced consistently stronger models than regression with census tracts on their own. This makes sense; most census tracts are very small, the median size of a census tract for this data being 0.85 square miles. When looking at a variable measuring something such as trips per person for a tract of that size, there may be a very small number of trips per person, yet at the same time a very high proportion of households with less vehicles than workers, because of bus trips in neighboring tracts. Looking at census tracts alone does not account for that, but grouping census tracts into the cities that they overlap with does.

The multiple linear regression model for census tracts used these variables: routes per square mile and proportion of land ¼ mile from a bus stop. It has an adjusted R^2 of 0.3964 and a p-value of $< 2.2e-16$. The equation was $\widehat{prophouseholds.lessvehicles} = 0.045348 + 0.001378(routes.per.sqmile) + 0.115716(prop.land.fourthmile.stop)$.

The multiple linear regression model for cities used these variables: routes per square mile, proportion of dwellings (homes) ¼ mile from a bus stop, and weekday departures (trips) per person. It had an adjusted R^2 of 0.7089 and a p-value of $1.178e-15$. The equation was $\widehat{prophouseholds.lessvehicles} = 0.0175730 + 0.0180655(routes.per.sqmile) + 0.0399738(propdwell.quarterm.stop) + 0.0006801(wkday.departures.per.route)$.

The cities model can account for 70.89% of the variance in proportion of households with less vehicles than workers in a city. Looking at the diagnostic plots, it does not violate linearity or normality, but does violate homoscedasticity, and demonstrates influential outliers.

These results are very promising. Going forward, I intend to adjust this model



and add more variables that measure access in terms of time span, route connections, and possibly better geographical access variables. With these three variables already accounting for 70.89% of the variance in the proportion of households with less vehicles than workers, it is very likely that additional measures of bus access would impact prediction and improve the overall strength of the model.

After creating these models, I went on to select cities with high residuals. All the predictors used in both models have a positive relationship with the target, so I selected cities with large positive residuals in my city regression model. These were specifically residuals over the 75th quantile. This produced these cities: Woodbine, Hillandale, Silver Spring, Redland, Brighton, Northwest Park / Oakview, Long Branch, Gaithersburg, Boyds, Germantown, Bethesda, Brookeville, Spencerville, Upper Rock Creek, and Burtonsville.

These are cities that have a higher proportion of households with less vehicles than workers than the model expects based on bus access variables. This implies that these may be areas where workers might lack a personal vehicle because they cannot have one (because of finances or some other reason). These are areas where bus service expansion may be especially likely to have a positive impact on the community, and where bus service expansion might have an especially large impact on the proportion of households with less vehicles than workers.

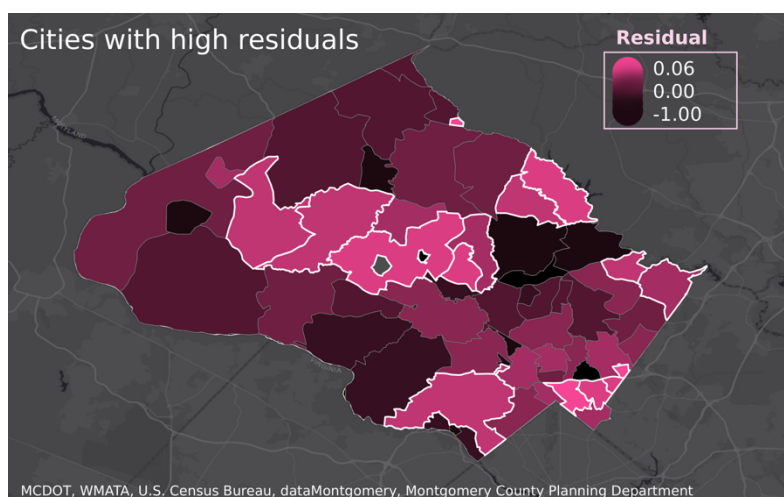


Figure 6: Cities with large positive residuals

Conclusion

It is clear from my analysis that bus service in Montgomery County is focused on areas where people live and is responding to need. Geographical bus access and bus access in terms of trips clearly correlates to and can contribute to prediction of households with less vehicles than workers in a city. Incorporating more measures of access, namely relating to time span, would likely impact prediction and improve linear regression models.

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